

Lab 3: Component Modelling & Architectural Pattern Selection

Centralized Audit Trail Compliance Engine

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1. Architectural Style Analysis

The three architectural styles considered are Layered, Microservices and Client-Server. Layered architecture provides separation of concerns but may introduce coupling and make individual scaling difficult. Client-Server provides centralized control and simple data consistency, but the central server can become a scalability bottleneck. Microservices divides the system into smaller independently deployable services, supporting independent scaling and fault isolation, although it introduces operational complexity, network communication and data-consistency challenges.

2. Architecture Selection

We chose **Microservices Architecture** for the Centralized Audit Trail Compliance Engine. The system has separate responsibilities for audit-log ingestion, validation, PII masking, storage, querying, export generation and cryptographic integrity verification. These responsibilities can be represented as independent components with defined interfaces.

3. Two Scenario-Related Reasons

Reason 1 — Performance requirement: The requirements specify complex filtering across up to 10 million historical audit records in under 1 second. A separate Audit Query & Filter Service can be optimized for this workload.

Reason 2 — Separation of responsibilities: Ingestion, validation, masking, storage, querying, export and verification perform different functions. Separating them gives each service a clear responsibility and supports independent maintenance.

4. Security Advantage

Authentication & Authorization is represented as a dedicated service for controlling access by the Compliance Officer and Security Auditor. The PII Detection & Masking Service is placed before persistent storage so sensitive fields are masked before records are stored. This supports the requirement that cleartext sensitive information must not be stored.

5. Performance Benefit

The Audit Query & Filter Service can be independently optimized and scaled for historical searches. This supports the stated enterprise-scale query performance requirement without requiring every service to scale at the same rate.

6. Component Diagram

The UML component diagram contains eight functional components: Authentication & Authorization Service, Audit Log Ingestion Service, Audit Validation Service, PII Detection & Masking Service, Audit Storage Service, Audit Query & Filter Service, Audit Export Service, and SHA-256 Integrity & Verification Service. The diagram also shows the three actors, provided interfaces using ball notation, required interfaces using socket notation, ports, solid internal service/data-flow arrows, and dashed <<use>> dependencies.

7. Alignment with the Problem Statement

The component design supports the previously identified functions: ingesting structured audit logs, parsing and masking PII before storage, searching and filtering historical audit records, generating immutable audit exports, computing SHA-256 integrity information, and allowing authorized security auditors to verify exported evidence.